

for free a three dimensional digital copy of the object and print it at home. The fact that virtually any substance in a powder form can be melted and fused to take on any shape or size could phase out moulding and casting processes of manufacturing.

With 3D printing, manufacturers can produce completely build products. This would not only reduce costs by minimising human intervention, it would also cut down on the time and energy required in assembling the product.

Further, as the part is produced by adding layer by layer as opposed to traditional machining methods where one is required to cut away material, the amount of wastage is almost negligible thus greatly reducing material costs.

I don't remember reading about it in the 90s

If you are wondering why a revolution in 3D printing is picking up just now and remained quiet through the 80's, 90's and 00's, it is because only in the second decade of the 21st century, that 3D printing became more accessible due to the fact that a lot of the technology is patented and are expiring one by one.

With 3D printing now in the hands of the general public, the entire DIY community embraced it like a long lost twin. It allowed the average Joe to tackle problems both at work and at home and also allowed these 'makers' to create new things.

3D printing is also a tool of empowerment for start up design and engineering firms. Instead of spending years working on a product, developing it and getting it production ready, with the help of a 3D printer, the same product could be brought in to the market in a matter of days.

Another reason why 3D printing didn't really catch on in the early days was due to the fact that the quality of raw materials was really bad. The finished prototype that the designer would print after designing the object using computer aided design (CAD), was not nearly as robust enough to be proud of the results. With time, the quality of the raw materials improved and blends of different plastics began to be used, yielding truly impressive results.



A model designed using CAD ready for print